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0.1 Exercise

Exercise 0.1.1 ► Calculate

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove in two different ways that for every $(\alpha, \beta) \in \Lambda^p V^* \times \Lambda^q V^*$,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

One way is to use definition, the other is to calculate in basis using the next exercise.

Proof. 1.

$$\alpha \wedge \beta = \frac{(p+q)!}{p!q!} \text{Alt}(\alpha \otimes \beta)$$

$$\beta \wedge \alpha = \frac{(p+q)!}{p!q!} \text{Alt}(\beta \otimes \alpha)$$

Only need to show that

$$\text{Alt}(\alpha \otimes \beta) = (-1)^{pq} (\beta \otimes \alpha)$$

Where

$$\text{Alt}(\alpha \otimes \beta) = \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) \sigma(\alpha \otimes \beta)$$

Let $\tau = (p + 1, \dots, p + q, 1, 2, \dots, p)$, then $\text{sgn}(\tau) = (-1)^{pq}$. We have

$$\begin{aligned}
 \text{Alt}(\alpha \otimes \beta) &= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) \sigma \circ \tau (\beta \otimes \alpha) \\
 &= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\text{sgn} \sigma) (\text{sgn} \tau) (\sigma \circ \tau) (\beta \otimes \alpha) \\
 &= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma' \in S_{p+q}} (\text{sgn} \sigma') \sigma' (\beta \otimes \alpha) \\
 &= (-1)^{pq} \text{Alt}(\beta \otimes \alpha)
 \end{aligned}$$

2. Suppose

$$\begin{aligned}
 \alpha &= \sum_{i_1 \dots i_p} a_{i_1 \dots i_p} e_{i_1}^* \wedge \dots \wedge e_{i_p}^* \\
 \beta &= \sum_{j_1 \dots j_q} b_{j_1 \dots j_q} e_{j_1}^* \wedge \dots \wedge e_{j_q}^*
 \end{aligned}$$

Then

$$\begin{aligned}
 \alpha \wedge \beta &= \sum_{i_1 \dots i_p} a_{i_1 \dots i_p} b_{j_1 \dots j_q} (e^*)_I \wedge (e^*)_J = \sum_{I,J} a_I b_J e_{IJ} \\
 \beta \wedge \alpha &= \sum_{I,J} a_I b_J e_{JI} = (-1)^{pq} \sum_{I,J} a_I b_J e_{IJ} = (-1)^{pq} \alpha \wedge \beta
 \end{aligned}$$

□

Exercise 0.1.2

Let V be a vector space of finite dimension and let V^* denotes the dual space. Prove that if $(\theta_1, \dots, \theta_r)$ is a family of r covectors in V^* which are **linearly dependent**, then the following wedge product:

$$\theta_1 \wedge \dots \wedge \theta_r = 0.$$

Proof. Since $\theta_1 \wedge \dots \wedge \theta_r$ is multi-linear and alternative, take any vector $v^1, \dots, v^r$, we have

$$\theta_1 \wedge \dots \wedge \theta_r (v^1, \dots, v^r) = \det(\theta_i(v^j))_{ij}$$

Since $\theta_1, \dots, \theta_r$ are linealy dependent , the row vectors of the matrix $(\theta_i(v^j))_{ij}$ are as well. Thus $\det(\theta_i(v^j))_{ij} = 0$, and $\theta_1 \wedge \dots \wedge \theta_r(v^1, \dots, v^r) = 0$. Since $v^1, \dots, v^r$ are arbitrary , $\theta_1 \wedge \dots \wedge \theta_r = 0$. $\square$

Exercise 0.1.3 ► Basis of Grassmann algebra

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove that

$$(e_{i_1}^* \wedge \dots \wedge e_{i_k}^*)_{1 \leq i_1 < \dots < i_k \leq n}$$

forms a basis of $\Lambda^k V^*$.

Proof. For $\alpha \in \Lambda^k V^*$, and $I = (i_1, \dots, i_k)$. Define

$$\alpha_I := \alpha(e_{i_1}, \dots, e_{i_k})$$

If I has a reputed index , then $\alpha_I = 0$, and if $J = \sigma I$, $\alpha_J = (\text{sgn} \sigma) \alpha_I$. We have

$$\sum_I' \alpha_I e_I^*(e_{j_1}, \dots, e_{j_k}) = \sum_I' \alpha_I \delta_J^I = \alpha_J = \alpha(e_{j_1}, \dots, e_{j_k})$$

Thus $\alpha = \sum_I' \alpha_I e_I^*$. Which shows that $e_{i_1}^*, \dots, e_{i_k}^*, 1 \leq i_1 < \dots < i_k \leq n$ spans the $\Lambda^k V^*$. To show that they are linealy independent. Suppose that

$$\sum_I k_I e_I^* = 0$$

Both sides act on e_J , we have $k_J = 0$. $\square$

Exercise 0.1.4 ► Grassmann algebra versus Polynomial functions on V .

Let V be a vector space of finite dimension and let V^* denotes the dual space. A p-multilinear map α on V is **symmetric** if

$$\alpha(v_1, \dots, v_n) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(n)})$$

for all permutations σ . Such α defines a homogeneous polynomial $\tilde{\alpha}$

defined by

$$\tilde{\alpha}(v) = \alpha(v, \dots, v)$$

for all $v \in V$. Then show that for symmetric multilinear maps α, β on V of respective p, q , the product $\alpha \circ \beta$ defined by

$$\alpha \circ \beta(v_1, \dots, v_{p+q}) := \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

is valued into symmetric multilinear map on V of degree $p + q$ and that

$$\alpha \circ \beta(v, \dots, v) = \alpha(v, \dots, v) \beta(v, \dots, v).$$

Could you compare with $\Lambda^\bullet V^*, \wedge$?

Exercise 0.1.5 ► Basic important examples of smooth manifolds

Please prove that the following topological spaces have a smooth manifold structure :

- The Euclidean sphere $S^n \subset \mathbb{R}^{n+1}$.
- The matrix groups $GL_n(\mathbb{R}), SL_n(\mathbb{R}), O_n(\mathbb{R})$, these are Lie groups of matrices. Each time could you tell me what is the dimension of these groups.

Proof. • Define F :

$$F(x^1, \dots, x^{n+1}) = x_1^2 + \dots + x_{n+1}^2$$

Then

$$S^n = F^{-1}(1)$$

$$dF_x = \nabla F(x) = (2x_1, \dots, 2x_{n+1}) \neq 0, \forall x \in \mathbb{R}^{n+1}$$

By regular value theorem, $S^n \subseteq \mathbb{R}^{n+1}$ is a regular embedded submanifold of $\mathbb{R}^{n+1}$, which itself is a manifold.

- 1. $\text{Mat}(n, \mathbb{R}) \simeq \mathbb{R}^{n^2}$ 是 n^2 维流形.
- 2. $GL(n, \mathbb{R}) = \{M \in \text{Mat}(n, \mathbb{R}) : \det M \neq 0\}$ 作为 $\text{Mat}(n, \mathbb{R})$ 的一个开子集也是一个 n^2 维流形.

3. $SL(n, \mathbb{R}) = \{M \in \text{Mat}(n, \mathbb{R}) : \det M = 1\}$

$$\begin{aligned}\det : \text{Mat}(n, \mathbb{R}) &\rightarrow \mathbb{R} \\ M &\mapsto \det M\end{aligned}$$

$SL(n, \mathbb{R}) = \det^{-1}(1)$. 取一个 $M \in \text{Mat}(n, \mathbb{R})$, $\det M = 1$. 要证明 $(d(\det))_M : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$. 按分量求偏导, 得到

$$\frac{\partial}{\partial a_{ij}} (\det M) = A_{ij} (\text{代数余子式})$$

由于 $\det M = 1$. 至少有一个 $A_{ij} \neq 0$. 从而 $(d(\det))_M$ 是满的.

4. 定义 $P : \text{Mat}(n, \mathbb{R}) \rightarrow \text{SMat}(n, \mathbb{R})$ (对称矩阵) $\simeq \mathbb{R}^{\frac{n(n+1)}{2}}$.

$$P(A) := A^T A$$

则 $O(n, \mathbb{R}) = P^{-1}(0)$. 对于 $A \in O(n, \mathbb{R})$.

$$(dP)_A : \text{Mat}(n, \mathbb{R}) \rightarrow \text{SMat}(n, \mathbb{R})$$

与上例不同, 这里用分量的形式去计算 Jacobi 是看不清楚的, 不过由于 P 本身足够简单, 我们可以用微分最原始的含义来观察. 考虑

$$P(A + \varepsilon B) - P(A) = \varepsilon (dP)_A(B) + O(\dots)$$

带入得到

$$(A^T + \varepsilon B^T)(A + \varepsilon B) - A^T A = \varepsilon (B^T A + A^T B) + O(\varepsilon)$$

于是 $(dP)_A(B) = B^T A + A^T B$. 只需要证明这个关于 B 的映射是满的. 为此, 解方程

$$A^T B + B^T A = C$$

取 $B = \frac{1}{2}AC$, 利用 A, C 对称, 带入得到

$$\frac{1}{2}A^T AC + \frac{1}{2}C^T A^T A = \frac{1}{2}(C + C^T) = C$$

故 $O(n, \mathbb{R})$ 是 $\text{Mat}(n, \mathbb{R})$ 的一个 $\frac{n(n-1)}{2}$ 维正则子流形. 注意到对于 $A \in O(n, \mathbb{R})$, $\det A = 1$ 或 $\det A = -1$, 可以窥见事实上 $O(n, \mathbb{R})$ 有两个连通分支.

□

Exercise 0.1.6 ► Quotient manifolds

Prove that the following objects are quotient manifolds

- $\mathbb{R}P^n$ obtained by quotienting S^n by antipodal map, show that it is nonorientable by a picture. Could you explain by a picture that the neighborhood of the line at infinity is homeomorphic to a Möbius ribbon ?
- Prove that the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is a smooth manifold.

Proof.

- 1. See $\mathbb{R}P^n$ is $S^2/(x \sim -x)$. The orientation changes at the equator. The map $A : S^2 \rightarrow S^2, A(x) = -x$ induces an identity map on $\mathbb{R}P^2$, dA send an orientation basis $\{e_1, e_2\}$ to one opposite to it .
- 2. Use $\mathbb{R}P^2 = D^2/\{x \sim -x \text{ for } x \in \partial D^2 = S^1\}$. Then the neighborhood of the line at infinity is a thin ring near ∂D^2 with some quotient relation. The quotient is just the same as that of Möbius .

□

Exercise 0.1.7 ► Vector fields are derivation

Let M be a C^∞ manifold. Show that the left C^∞ module of derivation of the ring $C^\infty(M)$ i.e. linear maps $D : C^\infty(M) \rightarrow C^\infty(M)$ satisfying

$$D(fg) = (Df)g + f(Dg)$$

can be identified with vector fields on M , smooth sections of TM .

Proof. Define

$$D(x)f = (Df)(x)$$

Then

$$D(x)(fg) =$$

□